"Open source enthusiasts can secure their career in the industry by making code and other contributions to open source projects, such as Linux kernel development..."

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Q What have you observed about the adoption of Linux and open source in the overall market? What's your take on the reservations some firms have regarding Linux and adopting open source—that these are risky for running mission/business-critical applications?

RB: I do not see a risk in using Linux for mission-critical projects. However, customers are divided in two different sets. One comprises the early technology adopters, who are the first to take risks. If everything works fine, they enjoy a longer life cycle with the product, and keep updating and innovating when better technologies come up. On the other hand, there are late adopters, who trust products only after seeking endorsement from various case studies, and are not interested in regular upgrades or experimentation, as long as the product runs fine.

Q What has your experience been with Linux and open source?

RB: I started my career with SVR 3.2 and SVR4 UNIX and used it on a multi-CPU kernel. The stability of Linux now is much better compared to how it was about five years ago. We know for sure that it is only going to get stronger in the years to come—not only because of the commitment of the open source community, but also due to the engagement of various commercial Linux vendors and hardware vendors.

The factory shipment of Windows OS is still higher, but we can't precisely estimate how many customers use Windows or Linux on Dell's PowerEdge servers, because various flavours of Linux are freely available, and customers are free to install and run them. Our key objective is for our hardware to run fine with the majority of them.

All of us, as IT professionals, have the technical knowledge and experience to solve our customers' pain points. The open source community is a great forum to collaborate with, to find solutions together, and fuel some great innovations. Let us commit to give back as much free software as we consume.

Q Do you offer any India-centric open source products?

RB: We currently have no open source-centric products targeted towards the Indian market. However, at Dell, we regularly interact with customers in India. Maybe there are specific needs within Linux for Indian customers, to effectively solve issues related to electricity voltage fluctuations, dusty environments, etc. Localisation of operating systems, which is very relevant to India, is taken care of by our Linux partners.

Q How does Dell contribute to open source?

RB: The Dell India R&D Centre team contributes to different open source programs through code submission. We propose enhancements to different subsystems of the operating system, and also decide how it will be implemented. For example, to solve an industry-wide problem of the NIC naming mechanism in enterprise Linux, our engineers conceptualised the new network device-naming convention. We worked with the Linux community to make sure that it gets completely accepted by them and becomes a part of the mainline upstream Linux kernel.

This 'Biosdevname' solution, proposed and owned by Dell, created an industry-standard method to name network cards based on their physical location in the server, rather than a generic sequential order. The project included the creation of a new ACPI standard and SMBIOS extensions, and is accepted in the mainline Linux kernel. As of today, this solution has been included in new versions of Red Hat, SuSE, etc.

We have been contributing in terms of various subsystems including networks, device drivers, systems management features, etc. Dell engineers and technologists from the India and Austin R&D centres have submitted a variety of code patches upstream, and contributed to specification changes for ACPI and IPMI.

JR: It is just not the number of lines, but the complexity of the code that matters more. Some of our engineers actively look at code, and have suggestions for